

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, SQL, C++, Bash

Tools/Frameworks: Docker, Kubernetes, AWS, GCP, Azure, FastAPI, Flask, TensorFlow, PyTorch, Databricks, Airflow, Kafka, Snowflake, Spark, dbt

Competence: Machine Learning, MLOps, Data Engineering, ETL Pipelines, Cloud Platforms, Model Deployment, Data Visualization, AI Architectures

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Developed and implemented physics-informed hybrid loss functions, enhancing model accuracy and stability.
- Collaborated with PhD researchers for a publication at the FLOMEKO conference.
- Built interactive dashboards for real-time data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased processing speed by 5x using advanced AI algorithms.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute cost and time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure.
- Built POCs for clients using YOLO, few-shot learning, and CNNs, showcasing AI capabilities.
- Conducted educational sessions on machine learning and big data at universities, mentoring over 50 students.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack web application providing AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and large language models for enhanced user experience.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented web scrapers and ETL pipelines for structured and unstructured data processing.
- Automated data ingestion and cleaning workflows, enabling efficient client analytics use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Developed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation.
- Achieved a 4% increase in model accuracy across MF, CC, and BP namespaces through custom feature engineering.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Designed and implemented a real-time Bitcoin price forecasting model using LSTM networks, achieving high prediction accuracy.

Saaz: AI Self-Playing Chitrali Sitar

2023

Signal Processing, Music Generation

- Built the world's first AI-driven self-playing Chitrali Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI-driven music technology.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Focused on advanced machine learning, data engineering, and cloud computing technologies.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Graduated with honors; recognized on the Dean's List of Honor for academic excellence.
- Specialized in machine learning, data science, and software development.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python